

Course Code	21CSC529T	Course Name	INFERENTIAL STATISTICS	Course Category	C	PROFESSIONAL CORE	L	T	P	C
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Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Data Science and Business Systems	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	Understand the role of statistics and its models in data analytics
CLR-2:	Get to know the probability and its application on data analysis
CLR-3:	Gain knowledge on sampling and hypothesis
CLR-4:	Implement the various hypothesis tests
CLR-5:	Know about independence, variance and statistical simulations

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	Categorize various probably representation to understand the data.	-	3	1
CO-2:	Use appropriate probability models in problem space	2	1	-
CO-3:	Collect/make sample data and choose hypothesis choices	-	3	2
CO-4:	Apply/ perform hypothesis test to infer the insight of resultant data	1	-	2
CO-5:	Implement various statistical test to analyze the data	3	1	-

Module-1 - Introduction To Statistics	9 Hour
Role of statistics in Data science - Different types of data. - Understanding Random variable Numerical Variable Categorical Variable - Data Collection, Types of Sampling, Descriptive Statistics - Measure of Central Tendency Mean, Median and Mode - Measure of Dispersion Range, Quartiles, Standard Deviation, Variance - Distribution of Data Skewness and Kurtosis - Covariance and Correlation, Difference between covariance and correlation and its significance. T1: Building programs to work with the Python packages for Statistics T2: Building programs to work with Descriptive statistics in Python T3: Building programs to perform Correlation analysis in Python	
Module-2 - Probability	9 Hour
Permutation and Combination – Types of Probability - Rules for computing probability - Marginal probability Conditional probability - Bayes Theorem - Applications of Bayes Theorem – Problem-solving in Bayes theorem - Probability Distributions Binomial & Poisson - Normal Distribution. T4: Building programs to perform Probability computation in Python	

T5: Building programs work with Bayes theorem applications in Python T6: Building programs work with Plotting and analyzing the distribution in Python	
Module-3 - Estimation & Hypothesis Testing	9 Hour
Sampling Distribution & Central limit theorem, Point estimate, confidence interval - Hypothesis Testing Basics Null & Alternative Hypothesis formulation - Types of hypotheses - Hypothesis testing process - Errors in hypothesis testing – Type I and Type II - Power of test - Number of tails - Choice of test statistic - Examples. T7: Simulations of Central Limit Theorem T8: Building Python programs to use Hypothesis testing in Python T9: Case study in Python using Estimation & Hypothesis Testing	
Module-4 – Test of Mean and Proportion	9 Hour
Test of Mean - One sample Z Test - Test of Mean - One sample t-Test - Test of Mean – 2 samples - Independent and Dependent - 2 samples - Test of proportion - One sample (Z proportion test) – Two samples (Z proportion test) - Case Study - Categorical vs Continuous – More than 2 samples. T10: Building programs to implement z-test and t-test in Python T11: Building programs to implement one sample and two sample tests in Python T12: Case study in Python	
Module-5 - Hypothesis Testing	9 Hour
Hypothesis Testing – Analysis of Variance - One way ANOVA Two-way ANOVA, Categorical vs Categorical – Chi-Square, Test of the goodness of fit - Test of independence - Test of variance T13: Building programs to implement the Chi-square test in Python T14: Building programs to implement non-parametric tests in Python T15: Case Study in Python	

Learning Resources	1. Kathryn A Szabat David M. Levine, P. K. Viswanathan, David Stephan (2017). Business Statistics: A First Course, 7th Edition	5. Hastie, T., Friedman, J., & Tibshirani, R. (2018). The Elements of statistical learning: Data mining, inference, and prediction. New York: Springer.
	2. Haslwanter T. An introduction to statistics with python. With applications in the life sciences. Switzerland: Springer International Publishing. 2016.	6. Grus, J. (2019). Data Science from Scratch, 2nd Edition. Place of publication not identified: O'Reilly Media, Inc.
	3. Downey, A., & Green Tea Press. (2012). Think Bayes: Bayesian statistics made simple. Needham, Massachusetts: Green Tea Press.	7. McKinney, W. (2018). Python for data analysis: Data wrangling with pandas, NumPy, and IPython. O'Reilly Media, Inc.
	4. Montgomery, D.C. and Runger, G.C. (2011). Applied Statistics and Probability for Engineers, John Wiley & Sons, 5th Edition	8. Vanderplas, J. T. (2017). Python data science handbook: Essential tools for working with data. O'Reilly Media, Inc.
		9. https://www.openintro.org/book/os/
		10. https://greenteapress.com/wp/think-stats-2e/

Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	20%	-	20%	-	20%	-
Level 5	Evaluate	5%	-	5%	-	5%	-
Level 6	Create	5%	-	5%	-	5%	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. Dr. Abhinanda Sarkar, Academic Director, Great Learning.	1. Dr. Masilamani, IITDM, Kanchipuram	1. Dr. A. Shobanadevi, DSBS
2. Dr Srabashi Basu, Senior Faculty, GreatLearning		2. Dr. D. Hemavathi, DSBS